
Abstract

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Keywords:

1. Introduction

The rapid transition toward low-carbon energy systems has significantly increased the penetration of renewable generation in modern power systems. While renewable resources such as wind and solar energy contribute to reducing carbon emissions, their inherent variability and uncertainty pose substantial challenges to power system planning and operation. In particular, the spatial mismatch between renewable generation sites and major load centers requires extensive transmission expansion, while temporal fluctuations in renewable output lead to inefficient utilization of network capacity and increased system operating costs.

Transmission network expansion planning (TNEP) has long been a fundamental problem in power systems and operations research. Classical TNEP models focus on determining the optimal set of transmission lines to construct in order to satisfy demand growth at minimum cost, typically formulated as mixed-integer optimization problems under DC power flow approximations [1, 2]. Over the years, a large body of literature has developed solution techniques, including decomposition methods and heuristic algorithms, to address the computational challenges of large-scale TNEP problems [3, 4].

With the increasing integration of renewable energy, recent studies have extended TNEP models to explicitly account for uncertainty in generation. Stochastic programming approaches have been widely used to model multiple renewable scenarios and capture expected system performance [5, 6], while robust optimization methods have been proposed to ensure system feasibility under worst-case realizations [7, 8]. These approaches improve system reliability but often rely heavily on transmission expansion as the primary flexibility resource.

In parallel, energy storage systems have emerged as an effective solution for enhancing system flexibility. By shifting energy across time, storage can mitigate renewable intermittency, reduce peak demand, and alleviate network congestion. Recent research has begun to explore the integration of energy storage into expansion planning problems. For example, co-optimization frameworks have been proposed to jointly determine transmission investments and storage deployment, demonstrating the potential of storage to defer or replace costly transmission upgrades [9, 10].

Despite these advances, existing studies on transmission and storage co-planning still exhibit several limitations. First, many models adopt simplified operational representations and fail to capture detailed multi-period storage dynamics. Second, the interaction between transmission expansion and storage investment under realistic multi-scenario operations remains insufficiently explored. Third, important operational aspects such as renewable curtailment and load shedding are often neglected or treated in a simplified manner, limiting the practical applicability of existing models.

To address these gaps, this paper proposes a comprehensive co-optimization model for transmission network expansion and energy storage siting in power systems with high renewable penetration. The proposed model integrates investment decisions for transmission lines and battery energy storage systems (BESS) with multi-scenario, multi-period operational decisions, including generation dispatch, power flow, storage operation, renewable curtailment, and load shedding. The objective is to minimize the total annualized system cost, encompassing both investment and expected operating costs.

The main contributions of this paper are summarized as follows:

- We develop a unified mixed-integer linear programming (MILP) model that jointly optimizes transmission expansion and energy storage deployment under a realistic multi-scenario operational framework.
- We explicitly capture the temporal dynamics of energy storage through state-of-charge constraints, enabling accurate modeling of inter-temporal flexibility.
- We incorporate renewable curtailment and load shedding into the objective function, allowing the model to balance economic efficiency, system reliability, and renewable utilization.
- We analyze the trade-offs between transmission investment and storage capacity under budget constraints, providing insights into cost-effective planning strategies for renewable-dominated power systems.

The remainder of this paper is organized as follows. Section 2 presents the problem formulation and mathematical model. Section ?? describes the solution approach. Section ?? reports numerical results and analysis. Finally, Section ?? concludes the paper.

2. Formulation

2.1. Problem description

This paper addresses a joint transmission network expansion and energy storage siting problem arising in power systems with high renewable energy penetration. The planning context is as follows.

A regional transmission network comprises a set of buses interconnected by existing transmission lines. To accommodate growing renewable generation, the system planner must decide which candidate transmission lines to construct and where to install utility-scale battery energy storage systems (BESS), subject to a total investment budget. The objective is to minimize the sum of annualized investment costs and expected system operating costs across a set of representative operating scenarios.

Each candidate transmission line l has a known construction cost, thermal capacity, and electrical susceptance. If built, it obeys the DC power flow approximation relating line flow to the voltage angle difference at its terminal buses. Each candidate storage site h is associated with a host bus and an upper bound on installable energy capacity. If selected, the BESS operates with given charge and discharge efficiencies and a fixed power-to-energy ratio that links the power rating to the installed energy capacity.

System operations are modeled over a set of representative scenarios $\mathcal{S}$, each consisting of multiple time periods $\mathcal{T}$ that capture diurnal variations in demand and renewable availability. For each scenario, the planner dispatches conventional generators, curtails renewable output if necessary, and manages storage charge–discharge schedules to meet demand at minimum cost. Involuntary load shedding is permitted as a last resort at a high penalty cost. A cyclic state-of-charge constraint is imposed within each scenario to ensure long-term storage sustainability.

The resulting model is a mixed-integer linear program (MILP) whose binary variables govern investment decisions and whose continuous variables describe power flows, generation dispatch, storage operations, and load shedding across all scenarios and time periods.

2.2. Optimization model

Tables 1–3 define the notation used in the model.

For each candidate line $l \in \mathcal{L}_C$, the big- M constant used in the DC power flow linearization is defined as:

$$M_l = B_l \cdot \bar{\Delta}_\theta. \quad (1)$$

Table 1. Set definitions

Symbol	Definition
$\mathcal{B}$	Set of buses
$\mathcal{L}_0$	Set of existing transmission lines
$\mathcal{L}_C$	Set of candidate transmission lines
$\mathcal{L}$	Set of all lines, $\mathcal{L} = \mathcal{L}_0 \cup \mathcal{L}_C$
$\mathcal{G}$	Set of conventional generators
$\mathcal{R}$	Set of renewable generators
$\mathcal{H}$	Set of candidate energy storage sites ($\mathcal{H} \subseteq \mathcal{B}$)
$\mathcal{S}$	Set of representative operating scenarios
$\mathcal{T}$	Set of time periods within each scenario, $\mathcal{T} = \{1, \dots, T\}$
$\mathcal{G}_b$	Subset of conventional generators located at bus b
$\mathcal{R}_b$	Subset of renewable generators located at bus b
$\mathcal{H}_b$	Subset of candidate storage sites at bus b
$\delta^+(b)$	Set of lines originating from bus b
$\delta^-(b)$	Set of lines terminating at bus b

This value provides a valid upper bound on $|B_l(\theta_{o(l)} - \theta_{d(l)})|$ for any feasible voltage angle assignment, ensuring that the DC power flow constraints are non-binding whenever the corresponding line is not constructed.

$$\begin{aligned} \min \quad & \sum_{l \in \mathcal{L}_C} c_l^{\text{line}} y_l + \sum_{h \in \mathcal{H}} (c_h^{\text{fix}} z_h + c^{\text{cap}} E_h) \\ & + \sum_{s \in \mathcal{S}} \omega_s \sum_{t \in \mathcal{T}} \left[\sum_{g \in \mathcal{G}} c_g^{\text{gen}} p_{g,s,t} \Delta t \right. \\ & \left. + c^{\text{shed}} \sum_{b \in \mathcal{B}} d_{b,s,t}^{\text{shed}} \Delta t + c^{\text{curt}} \sum_{r \in \mathcal{R}} (\bar{W}_{r,s,t} - w_{r,s,t}) \Delta t \right] \quad (2) \end{aligned}$$

$$\begin{aligned} & \sum_{g \in \mathcal{G}_b} p_{g,s,t} + \sum_{r \in \mathcal{R}_b} w_{r,s,t} + \sum_{h \in \mathcal{H}_b} (e_{h,s,t}^+ - e_{h,s,t}^-) \\ & + d_{b,s,t}^{\text{shed}} - D_{b,s,t} = \sum_{l \in \delta^+(b)} f_{l,s,t} - \sum_{l \in \delta^-(b)} f_{l,s,t}, \\ & \forall b \in \mathcal{B}, s \in \mathcal{S}, t \in \mathcal{T} \quad (3) \end{aligned}$$

$$\begin{aligned} f_{l,s,t} &= B_l(\theta_{o(l),s,t} - \theta_{d(l),s,t}), \\ & \forall l \in \mathcal{L}_0, s \in \mathcal{S}, t \in \mathcal{T} \quad (4) \end{aligned}$$

$$\begin{aligned} f_{l,s,t} - B_l(\theta_{o(l),s,t} - \theta_{d(l),s,t}) &\leq M_l(1 - y_l), \\ & \forall l \in \mathcal{L}_C, s \in \mathcal{S}, t \in \mathcal{T} \quad (5) \end{aligned}$$

$$\begin{aligned} f_{l,s,t} - B_l(\theta_{o(l),s,t} - \theta_{d(l),s,t}) &\geq -M_l(1 - y_l), \\ & \forall l \in \mathcal{L}_C, s \in \mathcal{S}, t \in \mathcal{T} \quad (6) \end{aligned}$$

Table 2. Parameter definitions

Symbol	Definition
c_l^{line}	Annualized investment cost for candidate line l (\$)
c_h^{fix}	Fixed annualized cost for building storage at site h (\$)
c^{cap}	Annualized investment cost per unit energy capacity (\$/MWh)
$\bar{F}_l$	Thermal capacity of line l (MW)
B_l	Susceptance of line l (p.u.)
$\bar{P}_g$	Maximum output of conventional generator g (MW)
c_g^{gen}	Marginal generation cost of generator g (\$/MWh)
$\bar{W}_{r,s,t}$	Available renewable output of generator r in scenario s , period t (MW)
$D_{b,s,t}$	Demand at bus b in scenario s , period t (MW)
ω_s	Weight of scenario s (days/year)
Δt	Duration of each time period (hours)
c^{shed}	Involuntary load shedding penalty cost (\$/MWh)
c^{curt}	Renewable curtailment penalty cost (\$/MWh)
$\bar{E}_h$	Maximum installable energy capacity at site h (MWh)
$\bar{\rho}$	Power-to-energy ratio for BESS (MW/MWh)
η^c	Charging efficiency of BESS
η^d	Discharging efficiency of BESS
Γ	Total investment budget (\$)
$\bar{\Delta}\theta$	Maximum admissible voltage angle difference across any line (rad)
b_{ref}	Index of the reference (slack) bus

$$-\bar{F}_l \leq f_{l,s,t} \leq \bar{F}_l, \quad \forall l \in \mathcal{L}_0, s \in \mathcal{S}, t \in \mathcal{T} \quad (7)$$

$$-\bar{F}_l y_l \leq f_{l,s,t} \leq \bar{F}_l y_l, \quad \forall l \in \mathcal{L}_C, s \in \mathcal{S}, t \in \mathcal{T} \quad (8)$$

$$0 \leq p_{g,s,t} \leq \bar{P}_g, \quad \forall g \in \mathcal{G}, s \in \mathcal{S}, t \in \mathcal{T} \quad (9)$$

$$0 \leq w_{r,s,t} \leq \bar{W}_{r,s,t}, \quad \forall r \in \mathcal{R}, s \in \mathcal{S}, t \in \mathcal{T} \quad (10)$$

$$e_{h,s,t}^+ \leq \bar{\rho} E_h, \quad \forall h \in \mathcal{H}, s \in \mathcal{S}, t \in \mathcal{T} \quad (11)$$

$$e_{h,s,t}^- \leq \bar{\rho} E_h, \quad \forall h \in \mathcal{H}, s \in \mathcal{S}, t \in \mathcal{T} \quad (12)$$

$$q_{h,s,t} = q_{h,s,t-1} + \eta^c e_{h,s,t}^- \Delta t - \frac{e_{h,s,t}^+}{\eta^d} \Delta t, \quad \forall h \in \mathcal{H}, s \in \mathcal{S}, t \in \mathcal{T} \quad (13)$$

$$q_{h,s,t} \leq E_h, \quad \forall h \in \mathcal{H}, s \in \mathcal{S}, t \in \{0\} \cup \mathcal{T} \quad (14)$$

$$q_{h,s,0} = q_{h,s,T}, \quad \forall h \in \mathcal{H}, s \in \mathcal{S} \quad (15)$$

Table 3. Decision variable definitions

Symbol	Definition
y_l	Binary; equals 1 if candidate line l is constructed
z_h	Binary; equals 1 if storage is built at site h
E_h	Continuous; energy capacity installed at site h (MWh)
$f_{l,s,t}$	Continuous; power flow on line l in scenario s , period t (MW)
$p_{g,s,t}$	Continuous; output of conventional generator g (MW)
$w_{r,s,t}$	Continuous; dispatched output of renewable generator r (MW)
$\theta_{b,s,t}$	Continuous; voltage angle at bus b (rad)
$d_{b,s,t}^{\text{shed}}$	Continuous; involuntary load shedding at bus b (MW)
$e_{h,s,t}^+$	Continuous; storage discharge power at site h (MW)
$e_{h,s,t}^-$	Continuous; storage charge power at site h (MW)
$q_{h,s,t}$	Continuous; state of charge at site h at the end of period t (MWh)

$$E_h \leq \bar{E}_h z_h, \quad \forall h \in \mathcal{H} \quad (16)$$

$$\sum_{l \in \mathcal{L}_C} c_l^{\text{line}} y_l + \sum_{h \in \mathcal{H}} (c_h^{\text{fix}} z_h + c^{\text{cap}} E_h) \leq \Gamma \quad (17)$$

$$0 \leq d_{b,s,t}^{\text{shed}} \leq D_{b,s,t}, \quad \forall b \in \mathcal{B}, s \in \mathcal{S}, t \in \mathcal{T} \quad (18)$$

$$\theta_{b_{\text{ref}},s,t} = 0, \quad \forall s \in \mathcal{S}, t \in \mathcal{T} \quad (19)$$

$$y_l \in \{0, 1\}, \quad \forall l \in \mathcal{L}_C; \quad z_h \in \{0, 1\}, \quad \forall h \in \mathcal{H} \quad (20)$$

$$E_h \geq 0, \quad \forall h \in \mathcal{H}; \quad e_{h,s,t}^+, e_{h,s,t}^- \geq 0, \quad \forall h, s, t \quad (21)$$

$$q_{h,s,t} \geq 0, \quad \forall h, s, t; \quad \theta_{b,s,t} \text{ free}, \quad \forall b, s, t \quad (22)$$

The objective function (2) minimizes the total annualized system cost, which consists of three components: investment costs for constructing candidate transmission lines and installing BESS (including both a fixed site preparation cost and a variable capacity cost), expected generation dispatch costs weighted by the scenario frequency ω_s , and penalty costs for involuntary load shedding and renewable curtailment. The curtailment of renewable generator r in scenario s , period t is captured implicitly as the difference $\bar{W}_{r,s,t} - w_{r,s,t}$ between the available and dispatched renewable output. Constraint (3) enforces power balance at each bus: the sum of conventional generation, renewable dispatch, net storage injection (discharge minus charge), and load shedding, minus the local demand, must equal the net power outflow through all incident transmission lines. The two terms on the right-hand side account for flows on lines originating from and terminating at the bus, respectively. Constraint (4) imposes the standard DC power flow relationship on existing lines, equating the power flow to the product of susceptance and voltage angle difference at the terminal buses. Constraints (5)–(6) extend this relationship to candidate lines using a big- M formulation: when $y_l = 1$ the two inequalities reduce to the equality $f_{l,s,t} = B_l(\theta_{o(l),s,t} - \theta_{d(l),s,t})$, and when $y_l = 0$ the big- M constant defined in (1) renders both constraints non-binding, allowing the flow to be driven to zero by the thermal limit (8). The value $M_l = B_l \cdot \bar{\Delta}_\theta$ provides the tightest valid bound, since $\bar{\Delta}_\theta$ is the maximum admissible angle difference across any single line. Constraints (7) and (8) bound the power flow on existing and candidate lines by their thermal capacities. For candidate lines, the capacity is multiplied by the construction indicator y_l , forcing the flow to zero on lines that are not built. Combined with (5)–(6), these constraints fully enforce the on–off behavior of candidate line investment. Constraint (9) limits conventional generator output to its installed capacity, and (10) limits renewable

dispatch to the scenario- and period-dependent available output, implicitly allowing curtailment whenever full output is not economical or physically deliverable. Constraints (11)–(15) govern energy storage operations. Constraints (11) and (12) bound the discharge and charge power by the product of the power-to-energy ratio $\bar{\rho}$ and the installed energy capacity E_h , linking operational limits to the sizing decision. Constraint (13) tracks the state of charge across consecutive time periods, where the energy stored increases with charging (scaled by efficiency η^c) and decreases with discharging (scaled by the reciprocal of η^d); the factor Δt converts power to energy. Constraint (14) ensures the state of charge never exceeds the installed energy capacity, and (15) imposes a cyclic boundary condition requiring the state of charge at the beginning and end of each scenario to coincide, preventing artificial depletion of stored energy. Constraint (16) couples the continuous capacity variable E_h to the binary site selection variable z_h : if storage is not built at site h ($z_h = 0$), the energy capacity is forced to zero; otherwise it may take any value up to the site-specific upper limit $\bar{E}_h$. Constraint (17) imposes the total investment budget, jointly limiting expenditure on transmission lines and storage installations. Constraint (18) bounds involuntary load shedding at each bus by the local demand, and (19) fixes the voltage angle at the reference bus to zero, eliminating the rotational degree of freedom inherent in the DC power flow model. Constraints (20)–(22) specify variable domains.

A key feature of this formulation is the coupling between transmission and storage investment decisions through the power balance constraint (3): building a transmission line increases the network's ability to export surplus renewable generation to distant load centers, whereas installing storage allows excess energy to be absorbed locally and released during periods of high demand. The budget constraint (17) forces these two investment channels to compete, and the multi-scenario operating cost evaluation determines the cost-effective allocation between them.

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